

Photoreceptor Contributions to Brightness Perception Preregistration Document

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With support from Johnson and Johnson, the University of Pennsylvania will conduct a brightness contrast matching experiment to determine how different photoreceptor classes contributed to the perception of brightness. We are particularly interested in the possibility of contributions from the intrinsically photosensitive retinal ganglion cells (ipRGCs). The experiment is specifically designed to characterize the relative contributions of melanopsin and short-wavelength-sensitive (S) cone signals in order to better understand how short-wavelength light influences brightness perception.

Using custom display technology, each trial will present two side-by-side grating patterns in Maxwellian view and ask participants to determine which grating has the “stronger light-dark pattern.” One grating—which we will call the “reference” stimulus—has a set achromatic contrast and an added perturbation targeting one or more photoreceptor classes of interest. The spatial frequency and achromatic contrast intensity of the reference also vary depending on the condition. The other grating—the “test” stimulus—is an achromatic pattern whose contrast is controlled by a QUEST+ algorithm to identify the point of subjective equality between the two gratings. Given existing knowledge about the spectral sensitivity each photoreceptor class, these brightness determinations will inform us about the degree to which various photoreceptor-directed perturbations influence perceived brightness contrast.

Our target enrollment is 24 subjects in total, comprising 12 females and 12 males between the ages of 18 and 39, with normal color vision and visual acuity. Each subject will complete 60 trials per condition. There are 24 conditions in total, consisting of:

- 3 photoreceptor perturbation directions (melanopsin, S-cone, and LMS-cone directed)
- 4 spatial frequencies (0.25, 0.5, 1, 4 cpd)
- 2 contrast levels (low and high)

Each subject will complete all 60 trials per condition in each of three separate testing sessions.

A more detailed description of this experimental plan exists and the details will be provided at the time of publication of results.